

Lecture 1

Phase space, flows and Poisson brackets

Objectives. *The mechanics you already know, rewritten so that the next ten lectures can be stated at all. Nothing here is a new physical law: every equation is one you can already write in coordinates. What changes is what the equations are about — a manifold, a vector field on it, and an algebraic operation on functions — and by the end two things follow that coordinates hide. “Conserved quantity” and “symmetry” become the same sentence, as a one-line consequence of the bracket being antisymmetric; and commuting integrals generate commuting flows, which is the engine of the central theorem of [Lecture 4](#).*

1.1 Why a manifold

The pendulum’s phase space is a cylinder

A plane pendulum is described by an angle θ and its conjugate momentum p . It is tempting to say the phase space is $\mathbb{R}^2$, and for most purposes nothing goes wrong. But θ and $\theta + 2\pi$ are the *same configuration*, not two different ones, so the honest phase space is

$$\mathcal{M} = S^1 \times \mathbb{R}, \quad (1.1)$$

a cylinder: the circle of angles, times the line of momenta.

This is not pedantry, and it is worth seeing the difference immediately. Take the Hamiltonian in units where the length, mass and gravity are 1,

$$H(\theta, p) = \frac{1}{2}p^2 - \cos \theta, \quad (1.2)$$

and ask what the level set $H = E$ looks like for $E > 1$ — a pendulum swinging right over the top.

- *On $\mathbb{R}^2$:* two curves, $p = \pm\sqrt{2(E + \cos \theta)}$, each running off to $\theta = \pm\infty$ without ever meeting itself. The angle grows without bound, and the trajectory is not periodic — nor is it a bounded set.
- *On the cylinder:* each of the two becomes a *single closed curve*, winding once around. The motion is periodic, with a period you can compute — and [Lecture 5](#) will compute it.

The same disagreement runs the other way below the separatrix: for $-1 < E < 1$ the plane shows infinitely many congruent ovals, one around each point $(2\pi k, 0)$, where the cylinder shows one oval, which is the one physical orbit.

A theorem that says “the motion is periodic” cannot even be stated on the wrong space. [Lecture 4](#), whose whole content is that the motion runs on a torus, is unstateable without this vocabulary, and that is what §§ 1.1 and 1.2 are for.

Charts

The circle is the smallest space that cannot be covered by one coordinate. The angle θ is a perfectly good coordinate on S^1 minus a point, but carry it all the way round and it jumps by 2π ; and there is no way to choose the cut so that this does not happen somewhere. Two overlapping charts suffice (Figure 1.1), and the price of the second chart is the whole apparatus of a manifold.

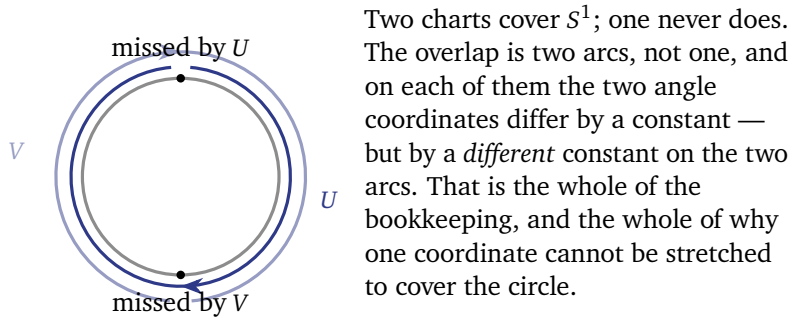


Figure 1.1: A chart on S^1 . The circle is the smallest example of a space that needs more than one coordinate patch, and the pendulum lives on it.

Definition 1.1 (Smooth manifold, in the form we need). A smooth manifold M of dimension m is a space covered by charts — one-to-one maps from open pieces of M onto open subsets of $\mathbb{R}^m$ — such that wherever two charts overlap, the map between their coordinates is smooth with smooth inverse. A function on M is smooth if it is smooth in every chart, which is well defined precisely because the transitions are.

Throughout Part I, M is the phase space of a mechanical system, and it is a manifold of dimension $m = 2n$, where n is the number of degrees of freedom. That is the only thing n ever means in this course; where a statement is about a general manifold and $2n$ would be misleading, the dimension is called m .

We shall not develop the general theory of manifolds — no embeddings, no partitions of unity, no orientability as a topic. Everything this course needs is: a space that looks like $\mathbb{R}^m$ in patches, on which it makes sense to differentiate.

Reading

Example 1.2 (Three phase spaces, and what they are not). • *Plane pendulum*: configuration space S^1 , phase space $T^*S^1 = S^1 \times \mathbb{R}$, dimension 2.

- *Double pendulum*: two angles, so configuration space is the torus $T^2 = S^1 \times S^1$, and phase space is $T^*T^2 = T^2 \times \mathbb{R}^2$, dimension 4. Emphatically not $\mathbb{R}^4$.
- *Free rigid body*: a configuration is an orientation, so configuration space is the rotation group $SO(3)$ and phase space is $T^*SO(3)$, dimension 6. We name it and go no further; $SO(3)$ is not simply connected, and that is a genuine physical fact — but it is not one this course needs.

Each of these is a *cotangent bundle*, “positions and their momenta”, and [Lecture 3](#) will show that every cotangent bundle carries the structure that makes Hamiltonian mechanics work, for free.

1.2 Tangent vectors, vector fields and flows

Two definitions of a tangent vector, and why both are needed

On the cylinder, what is a “velocity”? In $\mathbb{R}^m$ the answer is an arrow from one point to another, but on a curved space there is no arrow from a point to a distant point — only an infinitesimal one, at the point. The picture is Figure 1.2: a tangent vector at p is the velocity of a curve through p .

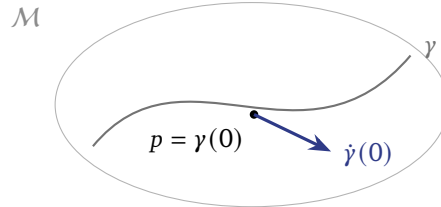


Figure 1.2: A tangent vector is the velocity of a curve through the point. This is the picture to keep; the definition that follows exists because a picture cannot be differentiated.

That picture is correct and it is the one to hold, but it cannot be computed with: two different curves through p can have the same velocity, and saying which pairs do requires coordinates and an equivalence relation. The computational definition avoids that by recording not the curve but what the curve *does to functions*.

Definition 1.3 (Tangent vector). A *tangent vector* at p is a linear map v from smooth functions near p to $\mathbb{R}$ satisfying the Leibniz rule

$$v(fg) = f(p) v(g) + g(p) v(f). \quad (1.3)$$

Such a map is called a *derivation* at p . The tangent vectors at p form a vector space $T_p\mathcal{M}$ of dimension m , and a chart $x^1, \dots, x^m$ gives it the basis $\partial/\partial x^1, \dots, \partial/\partial x^m$.

The two pictures are the same one. A curve γ with $\gamma(0) = p$ defines a derivation by

$$v(f) = \left. \frac{d}{dt} \right|_{t=0} f(\gamma(t)), \quad (1.4)$$

and every derivation arises this way. Read (1.4) from right to left and it says: *a tangent vector is a directional derivative*, which is exactly what a velocity is for. Keep Figure 1.2 as the picture and Definition 1.3 as the tool, and never expect either alone to do both jobs.

Vector fields, integral curves and flows

Definition 1.4 (Vector field, integral curve, flow). A *vector field* X assigns a tangent vector $X_p \in T_p\mathcal{M}$ to each point, smoothly; in a chart, $X = \sum_i X^i(x) \partial/\partial x^i$. An *integral curve* of X is a curve with $\dot{\gamma}(t) = X_{\gamma(t)}$. The *flow* φ_t sends a point p to $\gamma(t)$, where γ is the integral curve through p at $t = 0$.

In a chart the integral-curve condition is the system $\dot{x}^i = X^i(x)$: an ordinary differential equation, which is why the following is assumed rather than proved.

Theorem 1.5 (Existence and uniqueness · cited). Through each point of $\mathcal{M}$ there is exactly one integral curve of a smooth vector field X , defined for at least a short time, and it depends

smoothly on its starting point. Where both sides are defined, the flow satisfies $\varphi_{t+s} = \varphi_t \circ \varphi_s$ and $\varphi_0 = \text{id}$.

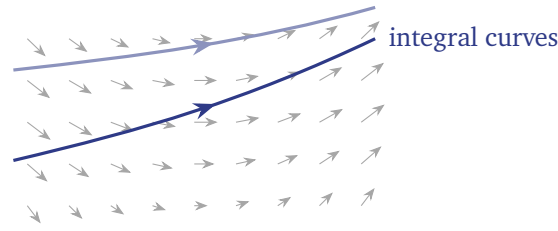


Figure 1.3: A vector field, and two of its integral curves. The arrows are given; the curves are what you get by following them. [Theorem 1.5](#) says exactly one curve passes through each point.

Reading

Near a point where $X \neq 0$ the flow can be straightened: there are coordinates in which $X = \partial/\partial x^1$ and the integral curves are parallel straight lines ([Figure 1.4](#)). Nothing local distinguishes one non-vanishing vector field from another — all the content is global, a theme that returns as Darboux's theorem in [Lecture 3](#).

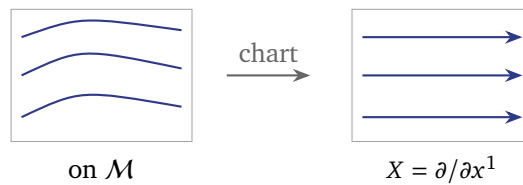


Figure 1.4: The flow box. Wherever $X \neq 0$ there is a chart in which the flow is translation at unit speed: a non-vanishing vector field has no local invariants at all.

Completeness, and why [Lecture 4](#) cares

[Theorem 1.5](#) promises an integral curve only for a short time. It may not last.

Definition 1.6 (Complete). X is *complete* if every integral curve is defined for all $t \in \mathbb{R}$ — equivalently, if φ_t is defined on all of $\mathcal{M}$ for every t .

Example 1.7 (An incomplete flow). On $\mathcal{M} = \mathbb{R}$ take $X = x^2 \partial/\partial x$, that is $\dot{x} = x^2$. Separating variables, the solution through $x_0 > 0$ is

$$x(t) = \frac{x_0}{1 - x_0 t}, \quad (1.5)$$

which escapes to infinity as $t \rightarrow 1/x_0$ and does not exist after that. Nothing is wrong with X : it is as smooth as could be wished. The particle simply leaves.

Proposition 1.8 (Compactness rescues completeness · sketched). A smooth vector field on a compact manifold is complete.

Proof (sketch). Suppose an integral curve exists on a maximal interval $[0, T)$ with T finite. The standard escape argument says such a curve must eventually leave every compact

subset — otherwise [Theorem 1.5](#), applied at a limit point, extends it past T and contradicts maximality. On a compact $\mathcal{M}$ there is nothing to leave. Hence $T = \infty$. ■

This looks like a technicality and it is not. In [Lecture 4](#) the proof of the Liouville–Arnold theorem needs the flows of n commuting integrals to be defined *for all time*, so that $\mathbb{R}^n$ — and not merely a small neighbourhood of 0 in $\mathbb{R}^n$ — acts on a level set. [Proposition 1.8](#) is where that comes from, applied to the level set rather than to $\mathcal{M}$, and it is exactly why the compactness hypothesis appears in the theorem. When [Lecture 9](#) finds a system whose level sets are *not* compact, this step has to be done by hand instead — and can be, there.

The Lie bracket

Two vector fields can be composed as differential operators, but XY is not a vector field: applied to a product it produces second derivatives and fails (1.3). The antisymmetric combination survives.

Proposition 1.9 (The Lie bracket). *For vector fields X, Y , the operator*

$$[X, Y] f = X(Yf) - Y(Xf) \quad (1.6)$$

is again a vector field, with components $[X, Y]^i = X(Y^i) - Y(X^i)$ in any chart.

Proof. Linearity is clear. For Leibniz, expand both terms on a product:

$$X(Y(fg)) = X(fYg + gYf) = fX(Yg) + gX(Yf) + (Xf)(Yg) + (Xg)(Yf),$$

and the same with X and Y exchanged. The two cross terms $(Xf)(Yg) + (Xg)(Yf)$ are *symmetric* under that exchange, so they cancel in the difference, leaving $[X, Y](fg) = f[X, Y]g + g[X, Y]f$. Those cross terms — note that they are products of *first* derivatives — are the obstruction to the Leibniz rule, and antisymmetry is exactly what removes them. The genuine second-derivative terms disappear separately and for a different reason: in coordinates they are $(X^i Y^j - Y^i X^j) \partial_i \partial_j f$, whose coefficient is antisymmetric in i, j while the Hessian is symmetric. The component formula follows by taking $f = x^i$. ■

What the bracket measures is the failure of the two flows to commute: follow X for time t , then Y for time s , then back along each, and the point you return to differs from where you started at order ts , by $[X, Y]$. We need only the extreme case, and we need it as an input to [Lecture 4](#).

Theorem 1.10 (Commuting fields, commuting flows · cited). *Two complete vector fields have commuting flows, $\varphi_t^X \circ \varphi_s^Y = \varphi_s^Y \circ \varphi_t^X$ for all s, t , if and only if $[X, Y] = 0$. (Arnold, Ch. 8.)*

1.3 Hamilton's equations as one vector field

Now specialise to $\mathcal{M} = T^*\mathbb{R}^n$ with coordinates $q_1, \dots, q_n, p_1, \dots, p_n$, and let H be a Hamiltonian. Hamilton's equations

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i} \quad (1.7)$$

are $2n$ scalar equations, and they are exactly the statement that the trajectory is an integral curve of the single vector field

$$X_H = \sum_{i=1}^n \left(\frac{\partial H}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial H}{\partial q_i} \frac{\partial}{\partial p_i} \right). \quad (1.8)$$

That is the whole of § 1.3: a mechanical system is a manifold with a vector field on it, and solving it means following the arrows.

One consequence is immediate and is worth drawing. Apply (1.8) to H itself: the two terms cancel in pairs, so $X_H H = 0$. Energy is conserved, and geometrically:

X_H is everywhere tangent to the level sets of H .

A trajectory can never leave the level set it starts on (Figure 1.5). That much is unconditional. How much more it buys depends on the level, and the distinction is worth stating now, because Lecture 4 builds a hypothesis out of it.

At a point where $dH \neq 0$ — equivalently, by (1.8), where $X_H \neq 0$ — the level set is locally a smooth surface of dimension $2n - 1$. At a *critical* point of H it need not be a surface at all.

- *One degree of freedom*, on a level where $dH \neq 0$: each connected component is a curve, and since X_H is a nowhere-zero field tangent to it, that component is a single orbit. This is the entire solution of the problem — but note both hypotheses. A level set with several components is several orbits, and a level through a critical point is not covered at all.
- $n \geq 2$: a single conserved quantity confines the motion to a $(2n - 1)$ -dimensional surface, again only where $dH \neq 0$, and no further. That is why Lecture 4 asks for n integrals rather than one — and why it asks for the level set to be *regular*.

The qualification is not hypothetical: the pendulum's level $E = 1$ in § 1.7 crosses itself at a critical point, and it is the lecture's principal worked example.

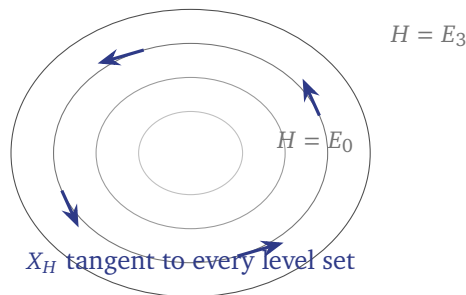


Figure 1.5: Level sets of H , with X_H tangent to them. Because $X_H H = 0$, a trajectory is trapped on the level set it starts on. For the harmonic oscillator these curves are the orbits, and the picture is the complete solution.

1.4 The Poisson bracket

Definition 1.11 (Poisson bracket). For smooth functions f, g on $T^*\mathbb{R}^n$,

$$\{f, g\} = \sum_{i=1}^n \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right), \quad \text{so} \quad \{q_i, p_j\} = \delta_{ij}. \quad (1.9)$$

Proposition 1.12 (The Poisson algebra). *The bracket is bilinear, antisymmetric $\{f, g\} = -\{g, f\}$, and satisfies the Jacobi identity and the Leibniz rule*

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0, \quad \{fg, h\} = f\{g, h\} + g\{f, h\}. \quad (1.10)$$

Proof. Bilinearity and antisymmetry are visible in (1.9), and Leibniz is the product rule applied to each derivative. Jacobi takes a line of bookkeeping, and it is worth doing here rather than deferring it, because § 1.5 proves the course's central identity from it.

Write $z^1, \dots, z^{2n}$ for all the coordinates at once and

$$\{u, v\} = \sum_{\alpha, \beta} \partial_\alpha u J^{\alpha\beta} \partial_\beta v, \quad J^{\alpha\beta} = \{z^\alpha, z^\beta\}, \quad (1.11)$$

so that J is a *constant* and *antisymmetric* matrix — that is all of (1.9) the argument uses. Expand the cyclic sum. Every term carries a second derivative of exactly one of f, g, h , so the terms fall into three classes, and it is enough to kill one: the others follow by relabelling f, g, h cyclically. Take the class with a second derivative of h . It comes only from the first two summands, since $\{h, \{f, g\}\}$ differentiates h just once, and it is

$$\underbrace{\partial_\alpha f J^{\alpha\beta} \partial_\gamma g J^{\gamma\delta} \partial_\beta \partial_\delta h}_{\text{from } \{f, \{g, h\}\}} + \underbrace{\partial_\alpha g J^{\alpha\beta} \partial_\beta \partial_\gamma h J^{\gamma\delta} \partial_\delta f}_{\text{from } \{g, \{h, f\}\}},$$

summed over repeated indices. Relabel the dummy indices of the second term by $\alpha \mapsto \gamma$, $\beta \mapsto \delta$, $\gamma \mapsto \beta$, $\delta \mapsto \alpha$: it becomes $\partial_\alpha f \partial_\gamma g \partial_\beta \partial_\delta h J^{\gamma\delta} J^{\beta\alpha}$. The two terms now differ only in carrying $J^{\alpha\beta}$ where the other carries $J^{\beta\alpha}$, so their sum has the factor $J^{\alpha\beta} + J^{\beta\alpha}$, which vanishes. The class is zero, and so are the other two. ■

Only two properties of J were used, that it is constant and antisymmetric, so the Jacobi identity has nothing to do with the particular pairing of each q with its own p . [Lecture 3](#) meets the same fact in geometric clothes: antisymmetry becomes ω being a 2-form, and constant coefficients make $d\omega = 0$ automatically. The converse fails — $\omega = e^q dq \wedge dp$ is closed without being constant — and it is *closedness*, not constancy, that corresponds to Jacobi in general.

The bracket is what turns § 1.3's geometry back into calculation. For any observable f , the chain rule along a trajectory together with (1.7) gives

$$\frac{df}{dt} = \sum_i \left(\frac{\partial f}{\partial q_i} \dot{q}_i + \frac{\partial f}{\partial p_i} \dot{p}_i \right) = \{f, H\}. \quad (1.12)$$

Definition 1.13 (First integral). f is a *first integral* — a conserved quantity — if $\{f, H\} = 0$.

So H generates time evolution, and (1.12) is the frozen convention of this course: $df/dt = \{f, H\}$, with the bracket in that order. Every sign in Part I follows from this choice together with (1.9), and § 1.5 is where that bill arrives.

Noether, in one line

The condition $\{f, H\} = 0$ is antisymmetric in its two arguments, so it vanishes for the pair (f, H) exactly when it vanishes for (H, f) . Reading (1.12) both ways:

$$f \text{ is conserved by the flow of } H \iff H \text{ is invariant under the flow of } f.$$

That is Noether's theorem, at this level of generality, and it is a one-line consequence of antisymmetry. A conserved quantity is a symmetry; they are not two facts related by a theorem but one fact read in two directions. Angular momentum is conserved because H is rotation-invariant, and — the same sentence — rotations are a symmetry because angular momentum is conserved.

1.5 From functions to vector fields: the engine

Equation (1.8) attached a vector field to H . Nothing there was special to H , and the general construction is the one the rest of Part I runs on.

Definition 1.14 (Hamiltonian vector field). For a smooth function f , the *Hamiltonian vector field* X_f is defined by its action on functions,

$$X_f h = \{h, f\}, \quad (1.13)$$

which in coordinates reads $X_f = \sum_i \left(\frac{\partial f}{\partial p_i} \partial_{q_i} - \frac{\partial f}{\partial q_i} \partial_{p_i} \right)$.

The order of arguments in (1.13) is fixed by wanting H to generate time evolution: taking $f = H$ gives $X_H h = \{h, H\} = dh/dt$ by (1.12), and the coordinate expression is (1.8). So $f \mapsto X_f$ is a map

$$\{\text{observables, with } \{\cdot, \cdot\}\} \longrightarrow \{\text{vector fields, with } [\cdot, \cdot]\},$$

from one bracket to another. The question this section answers is whether it respects them.

Theorem 1.15 (The engine). For all smooth f, g ,

$$X_{\{f, g\}} = -[X_f, X_g]. \quad (1.14)$$

We prove it twice. The first proof is a computation and shows the identity is true; the second is three lines and shows *why*.

First proof, in coordinates. Take $n = 1$; the general case is the same with sums over i . Write subscripts for partial derivatives, so $X_f = f_p \partial_q - f_q \partial_p$. For vector fields $A = a \partial_q + b \partial_p$ and $B = c \partial_q + d \partial_p$, Proposition 1.9 gives $[A, B] = (Ac - Ba) \partial_q + (Ad - Bb) \partial_p$. With $a = f_p$, $b = -f_q$, $c = g_p$, $d = -g_q$ the ∂_q component of $[X_f, X_g]$ is

$$X_f g_p - X_g f_p = (f_p g_{qp} - f_q g_{pp}) - (g_p f_{qp} - g_q f_{pp}).$$

On the other side, $\{f, g\} = f_q g_p - f_p g_q$, so the ∂_q component of $X_{\{f, g\}}$ is

$$\{f, g\}_p = f_{qp} g_p + f_q g_{pp} - f_{pp} g_q - f_p g_{qp},$$

which is the negative of the previous display, term for term. The ∂_p components are the identical computation and give

$$X_f(-g_q) - X_g(-f_q) = f_{qq} g_p + f_q g_{pq} - f_{pq} g_q - f_p g_{qq} = \{f, g\}_q,$$

against $-\{f, g\}_q$ for $X_{\{f, g\}}$ — again a sign apart. Hence $[X_f, X_g] = -X_{\{f, g\}}$. ■

Second proof, from the Jacobi identity. Apply both sides to an arbitrary function h and use only (1.13). The right-hand side is

$$[X_f, X_g] h = X_f \{h, g\} - X_g \{h, f\} = \{\{h, g\}, f\} - \{\{h, f\}, g\}.$$

The left-hand side is $\{h, \{f, g\}\}$, and the Jacobi identity (1.10), rearranged with antisymmetry, says exactly $\{h, \{f, g\}\} = \{\{h, f\}, g\} - \{\{h, g\}, f\}$. The two sides are therefore negatives of each other, for every h . ■

The second proof locates the content: *the engine is the Jacobi identity*, wearing different clothes. The map $f \mapsto X_f$ is an anti-homomorphism of Lie algebras, and it is one because the bracket satisfies Jacobi — which is also, as [Lecture 3](#) will show, the same statement as $d\omega = 0$.

The minus sign is not a typo

It will look like one, so here it is in the smallest possible case, with one degree of freedom and $f = q^2/2$, $g = p^2/2$:

$$\begin{aligned} X_f &= -q \partial_p, & X_g &= p \partial_q, & [X_f, X_g] &= -q \partial_q + p \partial_p, \\ \{f, g\} &= qp, & X_{qp} &= q \partial_q - p \partial_p, & X_{\{f, g\}} &= -[X_f, X_g]. \end{aligned} \quad (1.15)$$

Compute the five entries and the sign is unavoidable. It is forced by the physicists' convention $df/dt = \{f, H\}$ of (1.12), together with (1.13); sources that write $X_f h = \{f, h\}$, or that put the bracket the other way round in the evolution law, get a plus, and they are not making a different claim about mechanics. Our conventions are frozen in the course's notation bible and re-derived symbolically in `scripts/check-conventions.py`; they have been wrong here once, and the remedy was to check them rather than to argue about them.

What the course actually uses

Corollary 1.16 (Commuting integrals, commuting flows). *If $\{F_i, F_j\} = 0$ then $[X_{F_i}, X_{F_j}] = 0$, and if in addition the fields are complete, their flows commute.*

Proof. $X_{\{F_i, F_j\}} = X_0 = 0$, so [Theorem 1.15](#) gives $[X_{F_i}, X_{F_j}] = 0$; then apply [Theorem 1.10](#). ■

[Corollary 1.16](#) is the only use Part I ever makes of [Theorem 1.15](#), and it **does not see the sign**: zero is zero either way. That is worth saying plainly, because it tells you how much to worry about a minus sign that has just been laboured over — for our purposes, not at all. It is laboured over because it must be *printed* correctly, not because the course's conclusions depend on it.

What the corollary buys is large. It converts an algebraic hypothesis that can be *checked* — n functions whose brackets vanish — into a geometric conclusion, n commuting flows, and hence an action of the abelian group $\mathbb{R}^n$ on each level set. That action is the whole machine of the Liouville–Arnold theorem in [Lecture 4](#), and [Theorem 1.15](#) is what starts it.

1.6 Liouville's theorem on phase-space volume

Theorem 1.17 (Liouville · stated; proved on Sheet 1). *The flow of X_H preserves the phase-space volume $dq_1 \cdots dq_n dp_1 \cdots dp_n$: for any region R and any time t for which φ_t is defined on all of R , $\text{vol}(\varphi_t(R)) = \text{vol}(R)$.*

The qualification is not pedantry, and [Example 1.7](#) is the reason. Hamiltonian flows can be incomplete like any others: on $T^*\mathbb{R}$ the perfectly smooth $H = q^2 p$ gives $\dot{q} = q^2$, $\dot{p} = -2qp$, whose solutions $q(t) = q_0/(1 - q_0 t)$ escape in finite time, so φ_1 is not defined anywhere with

$q_0 \geq 1$ and $\text{vol}(\varphi_1(R))$ need not be a number at all. What the theorem says is that volume is preserved wherever the flow exists — and on a compact level set, [Proposition 1.8](#) says that is everywhere.

The reason is one line. A flow changes volume at a rate given by the divergence of its vector field, and from (1.8)

$$\text{div } X_H = \sum_i \left(\frac{\partial}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial}{\partial p_i} \frac{\partial H}{\partial q_i} \right) = 0 \quad (1.16)$$

by equality of mixed partials. Hamiltonian flow is incompressible. Note what is *not* claimed: regions keep their volume, but they need not keep their shape, and generically they are stretched and folded beyond recognition ([Figure 1.6](#)) — which is how a volume-preserving flow can look thoroughly chaotic, as [Lecture 6](#) will show that it does.

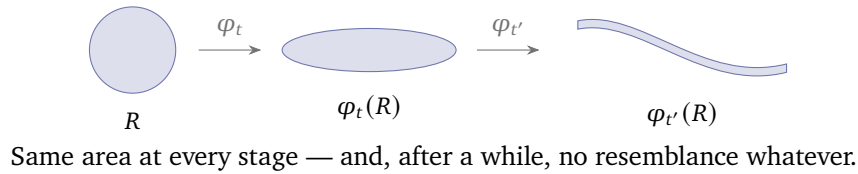


Figure 1.6: Liouville's theorem. The flow carries a region to regions of equal volume, and says nothing at all about their shape.

[Lecture 3](#) will restate [Theorem 1.17](#) in a single line, once the volume is recognised as $\omega^n/n!$ for the symplectic form ω , and [Lecture 6](#) will use it to argue that trajectories cannot simply settle down.

1.7 Worked phase portraits

For one degree of freedom [§ 1.3](#) has already solved every problem qualitatively: away from the critical points of H , the orbits are the connected components of its level curves, traversed in the direction of X_H . Two portraits are worth having by heart — and the second is interesting precisely where that qualification bites.

The harmonic oscillator, $H = \frac{1}{2}(p^2 + \nu^2 q^2)$. For $E > 0$ the level sets are ellipses about the origin, all closed, all traversed clockwise; the period is $2\pi/\nu$ and does not depend on which ellipse. That independence is special, and [Lecture 5](#) will see it break. The level $E = 0$ is the origin alone — the one critical point, an equilibrium and not an ellipse.

The pendulum, $H = \frac{1}{2}p^2 - \cos \theta$ on the cylinder (1.1), is the richer picture ([Figure 1.7](#)) and the course's standing example. Three regimes, separated by one curve:

- $-1 < E < 1$: *libration*. Closed curves encircling $(\theta, p) = (0, 0)$; the bob swings back and forth.
- $E = 1$: the *separatrix*. Here $p^2 = 2(1 + \cos \theta) = 4\cos^2(\theta/2)$, so $p = \pm 2\cos(\theta/2)$ for $\theta \in (-\pi, \pi)$ — the range the cylinder coordinate runs over, and the range on which $\cos(\theta/2) \geq 0$ makes the square root come out without a modulus. This level is the one place where the warning of [§ 1.3](#) bites: $\theta = \pi$, $p = 0$ is a critical point of H , so the level set is not a curve there — two branches cross — and it is *not one orbit*. It is three: the equilibrium itself, and the two curves that leave it and return to it, each taking infinitely long at both ends. Nothing on this level is periodic, though it is drawn as a closed curve on the page.

- $E > 1$: *rotation*. Curves winding around the cylinder, p never changing sign; the bob goes over the top for ever.

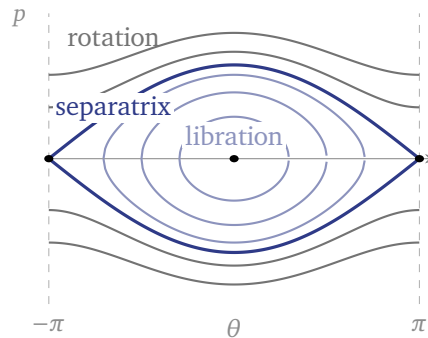


Figure 1.7: The pendulum's phase cylinder. The left and right edges are the same line: a rotation curve leaving on the right returns on the left, and is a *single closed orbit*. The separatrix (heavy) divides libration from rotation and belongs to neither. The three dots are equilibria — stable at $\theta = 0$, unstable at $\theta = \pm\pi$, which on the cylinder is *one* point, not two. Every later lecture that computes something for the pendulum computes it separately in the two regimes, because they are genuinely different curves.

The separatrix is where the two families of closed curves fail to be deformable into one another, and it is the first sign of something this course returns to repeatedly: the interesting structure of an integrable system sits at the places where its level sets change character. [Lecture 5](#) computes the period in both regimes and watches it diverge at the separatrix.

Summary.

- *Phase space is a manifold*. The pendulum's is the cylinder $S^1 \times \mathbb{R}$, not $\mathbb{R}^2$ (1.1), and on the cylinder a rotating pendulum has one periodic orbit rather than infinitely many curves.
- *Dynamics is a vector field*. Hamilton's equations are the single field X_H of (1.8); since $X_H H = 0$, trajectories are trapped on level sets of H .
- *Completeness*. Integral curves can run out of time ([Example 1.7](#)); on a compact manifold they cannot ([Proposition 1.8](#)). [Lecture 4](#) needs exactly this, on a level set.
- *The bracket*. $df/dt = \{f, H\}$ (1.12), and a first integral is $\{f, H\} = 0$. By antisymmetry alone, “ f is conserved by H ” and “ H is invariant under f ” are the same statement — Noether in one line.
- *The engine*. $X_{\{f,g\}} = -[X_f, X_g]$ ([Theorem 1.15](#)), which is the Jacobi identity in disguise. The minus is forced by $df/dt = \{f, H\}$ and is not a typo; and the only use Part I makes of the identity — commuting integrals give commuting flows, [Corollary 1.16](#) — cannot see it.
- *Liouville*. Hamiltonian flow preserves phase-space volume ([Theorem 1.17](#)), because X_H is divergence-free. Volume, not shape.

Tutorial. Exercise Sheet 1.

References

Arnold, *Mathematical Methods of Classical Mechanics*, Ch. 8 (Hamiltonian vector fields, the Lie algebra of Hamiltonian functions, Liouville's theorem); da Silva, *Lectures on Symplectic Geometry*, §§1–3 and 18 (manifolds, vector fields, Poisson structures); Babelon, Bernard and Talon, *Introduction to Classical Integrable Systems*, Ch. 1. **Sign conventions differ across all three sources for precisely the identity of § 1.5** — check any formula against (1.14) and (1.15) before transcribing it.